App objects

App objects reference

Reference information for building app objects, including configuration options for your project.

> This feature requires approval from HubSpot to use. If you're interested in applying to get access to app objects, or if you want to learn more about the functionality, please submit **this in-app form**.

Below, find reference information for developer platform app features with app objects, including configuration file definitions, scopes details, and more.

Project structure

All project components must live within the `src` directory specified in the `hsproject.json` config file.

All app features and components must live within the `app/` directory.

App objects are defined within the `app-objects/` directory.

App object associations are defined within the `app-object-associations/` directory.

All component and feature instances are declared using `*-hsmeta.json` files. You can use any file name you'd like, as long as it ends in `-hsmeta.json` (e.g., `my-cool-object-hsmeta.json`). These files must live at the root level of their respective folder.

```
project-folder/
└── src/
    └── app/
        ├── app-hsmeta.json
        └── app-objects/
            └── app-object-hsmeta.json
        └── app-object-associations/
            └── app-object-to-contact-hsmeta.json
```

HubSpot Developers

›

```
project-folder/
└──────src/
    └──── app/
        └──── app-hsmeta.json
        └──── app-objects/
            └──── my-cool-object-hsmeta.json
```

Below are the configuration options available for `*-object-hsmeta.json` .

HubSpot Developers

```
"name": "CAR",
"appPrefix": "Vroom",
"description": "An automobile in our warehouse.",
"singularForm": "Car",
"pluralForm": "Cars",
"primaryDisplayLabelPropertyName": "model",
"secondaryDisplayLabelPropertyNames": ["make"],
"settings": {
  "hasRecordPage": true,
  "allowsUserCreatedRecords": true,
  "hasEngagements": true
},
"properties": [
  {
    "type": "string",
    "fieldType": "text",
    "name": "model",
    "label": "Model",
    "description": "The model of the car"
  },
  {
    "type": "enumeration",
    "fieldType": "select",
    "name": "make",
    "label": "Make",
    "description": "The manufacturer of the car.",
    "options": [
      {
        "label": "Ford",
        "value": "ford",
        "displayOrder": 0
      },
      {
        "label": "Toyota",
        "value": "toyota",
        "displayOrder": 1
      },
      {
        "label": "Chevrolet",
```

HubSpot Developers

```
    ]
  }
}
```

Fields marked with * are required.

Field	Type	Description
`uid` *	String	A unique identifier for the app object. Must be globally unique within the project.
`type` *	String	The type of component. Must match the name of the parent folder (`app-object`).
`name` *	String	The name of your app object. Use the approved name that you received in your approved app confirmation. Must be uppercase snake case (`MY_OBJECT_NAME`).
`appPrefix`	String	A string that precedes the singular or plural name of the object in HubSpot's UI to help differentiate it from other objects. In this example, the `appPrefix` of `Vroom` and `singularForm` of `Car` would result in "Vroom Car" displaying in the UI.
`description` *	String	the description of the object, which will display in HubSpot.
`singularForm` *	String	The singular form of the object name.
`pluralForm` *	String	The plural form of the object name.
`properties` *	Array	A list of CRM properties defined for the object. Properties are defined using the same fields as the **properties API**. The resulting properties will automatically have `a<appId>_` prepended to it (e.g., `a12345_make`) on creation. You should not include the prefix in the config file.
`primaryDisplayLabelPropertyName` *	String	The name of the property that should be used as the **primary display property**. The value should match the name provided in the `properties` list (i.e., it should not include the generated `a<appId>_` prefix.)

HubSpot Developers

›

		prefix.)
`settings` *	Object	An object containing object settings.
`hasRecordPage` *	Boolean	Whether record pages exist for instances of this app object. When set to `false`, the index page will still exist, but it will not include links to individual record pages.
`allowsUserCreatedRecords` *	Boolean	Whether end-users will be able to create records using the object. When set to `false`, users will not be able to create records in HubSpot
`hasEngagements` *	Boolean	Whether the app object records support activities/engagements. When set to `false`, app object record pages will not include any engagement functionalities, such as the activity tab or activity-related timeline filters.

The fully-qualified name (FQN) for your app object will be `a<appId>_<objectName>`. For example: if your `appId` is `16858319` and the app object name is `CARS`, then the FQN would be `a16858319_CARS`. You'll use the FQN when setting scope values for your app objects.

App schema

To create an app object, include an `app-hsmeta.json` configuration file in the `app` directory.

```
project-folder/
└── src/
    └── app/
        └── app-hsmeta.json
```

Below are the configuration options available for `app-hsmeta.json`.

HubSpot Developers

```
    "description": "An example to demonstrate how to build an app with an app c
    "name": "my first app object app",
    "distribution": "marketplace",
    "auth": {
      "type": "oauth",
      "redirectUrls": ["http://localhost:3000/oauth-callback"],
      "requiredScopes": ["crm.objects.contacts.read", "crm.objects.contacts.wri
      "optionalScopes": [],
      "conditionallyRequiredScopes": []
    },
    "permittedUrls": {
      "fetch": ["https://api.hubapi.com"],
      "iframe": [],
      "img": []
    },
    "support": {
      "supportEmail": "support@example.com",
      "documentationUrl": "https://example.com/docs",
      "supportUrl": "https://example.com/support",
      "supportPhone": "+18005555555"
    }
  }
}
```

App schema *-hsmeta.json fields

Fields marked with * are required

Field	Type	Description
`uid` *	String	An internal unique identifier for the app. Must be globally unique within the project. Can be any string up to 64 characters. Characters can be uppercase or lowercase, and can include numbers, underscores (`_`), dashes (`-`), and periods (`.`).
`type` *	String	The type of component. Must match the name of the parent folder (`app`).

HubSpot Developers

›

		be any string up to 200 characters. Must not start or end with a whitespace character.
`distribution` *	String	The method of app distribution. Must be set to `marketplace` for the app to be eligible for HubSpot Marketplace listing.
`auth` *	Object	An object containing the app's authentication method details. See the **auth table** for details.
`permittedUrls`	Object	An array containing the URLs that the app is allowed to call. URLs must use the HTTPS scheme and must contain an **authority**, followed by an optional path prefix if needed.
`supportEmail`	String	A valid email address that users can contact for support.
`documentationUrl`	String	The external URL that users can navigate to for supporting documentation. Must use HTTPS.
`supportUrl`	String	The external URL that users can navigate to for additional support. Must use HTTPS.
`supportPhone`	String	The phone number that users can contact for support. Must start with a plus sign (`+`).

auth fields

Fields marked with * are required.

Field	Type	Description
`type` *	String	The type of authentication. Must be set to `oauth` for the app to use OAuth authentication.
`redirectUrls` *	Array	A list of URLs that the OAuth process is allowed to reroute back to. Each app must have at least one auth redirect URL, and it must use HTTPS. The only exception is that `http://localhost` is allowed for testing.
`requiredScopes` *	Array	A list of your app's required scopes. Each app must include at least one scope, and the installing user must

HubSpot Developers

		be excluded from the authorization during installation if the account or user installing the app doesn't have the proper permissions. In that case, the scope will not be included in the resulting refresh token or access token. **Learn more about scopes below**.
`conditionallyRequiredScopes`	Array	A list of scopes that are required only when included in the `scope` query parameter of the install URL. **Learn more about scopes below**.

Scopes

In the `auth` field of an app configuration file, you can specify three **types of scopes**: required scopes, conditionally required scopes, and optional scopes. If you're just getting started with app objects, you should only include your app object scopes as `conditionallyRequiredScopes`. This will allow you to silo your new features to specific customers by including the app object scopes in the install URL.

App object scopes use the following format:

```
crm.app.schemas.<appObjectFullyQualifiedName>.read
```

For example, for an app object with the FQN `a16858319_cars`, the `read` scope would be:

```
crm.app.schemas.a16858319_cars.read
```

At a minimum, your app must include the above `read` scope to enable customers to access the object. It's recommended to include all app object scopes in your app, as shown below.

HubSpot Developers

```
        "crm.objects.contacts.read",
        "crm.objects.contacts.write",
        "crm.app.objects.a12345_MY_APP_OBJECT.view",
        "crm.app.objects.a12345_MY_APP_OBJECT.create",
        "crm.app.objects.a12345_MY_APP_OBJECT.edit",
        "crm.app.schemas.a12345_MY_APP_OBJECT.read",
        "crm.app.objects.a12345_MY_APP_OBJECT.merge",
        "crm.app.objects.a12345_MY_APP_OBJECT.delete",
        "crm.app.schemas.a12345_MY_APP_OBJECT.properties.write"
      ],
      "optionalScopes": [],
      "conditionallyRequiredScopes": []
    },
```

For a full list of available scopes, see the .

Webhooks component definition

To define a set of webhook subscriptions for your app, include a `webhooks` directory in the project, along with a `*-hsmeta.json` configuration file.

```
project-folder/
└── src/
    └── app/
        ├── app-hsmeta.json
        └── webhooks/
            └── webhook-hsmeta.json
```

Below are the available configuration options for the `*-hsmeta.json` file.

HubSpot Developers

```
    "settings": {
      "targetUrl": "https://example.com/webhook",
      "maxConcurrentRequests": 10
    },
    "subscriptions": {
      "crmObjects": [
        {
          "subscriptionType": "object.creation",
          "objectType": "contact",
          "active": true
        },
        {
          "subscriptionType": "object.propertyChange",
          "objectType": "car_app_object",
          "propertyName": "carProperty",
          "active": true
        }
      ],
      "legacyCrmObjects": [
        {
          "subscriptionType": "contact.propertyChange",
          "propertyName": "lastname",
          "active": true
        },
        {
          "subscriptionType": "contact.deletion",
          "active": true
        }
      ],
      "hubEvents": [
        {
          "subscriptionType": "contact.privacyDeletion",
          "active": true
        }
      ]
    }
  }
}
```

HubSpot Developers

`uid` *	String	An internal unique identifier for the webhook component.
`type` *	String	The type of component, which should be `webhooks` in this case.
`settings` *	Object	An object that specifies two fields: `targetUrl`, which is the publicly available URL for HubSpot to call where event payloads will be delivered, and `maxConcurrentRequests`, which is the upper threshold of HTTP requests that HubSpot will make in a given time frame.
`subscriptions` *	Object	An object that specifies the subscription types your app will subscribe to.
`crmObjects`	Array	An array containing event subscription definitions. This is the standard array to include, and should be used for all events in the new format (`object.*`). Classic webhook subscription types should instead be included in `legacyCrmObjects` and `hubEvents` arrays, depending on the event.
`legacyCrmObjects`	Array	An array containing classic subscription types, such as `contact.creation` and `deal.deletion`.
`hubEvents`	Array	An array containing the classic subscription types `contact.privacyDeletion` and `conversation.*`

Please note: the `associationChange` event is not currently supported for app objects.

For each `subscription` object, the following fields can be specified, based on the subscription definition type you're subscribed to (i.e., `crmObjects`, `legacyCrmObjects`, or `hubEvents`) or whether you're subscribing to a specific property change (e.g., `contact.propertyChange`).

Field	Type	Description
`subscriptionType`	String	The type of event being subscribed to.

HubSpot Developers

›

		car_app_object).
propertyName	String	For property change subscriptions, this specifies which property will trigger the webhook event.
active	Boolean	Whether webhook events will be triggered for this subscription.

App card schema

To create an app card that appears on an app object record page, include a `cards` component directory in the project, along with a configuration file.

```
project-folder/
└── src/
    └── app/
        ├── app-hsmeta.json
        └── cards/
            └── my-app-card-hsmeta.json
            └── my-app-card.jsx
```

Make sure you've run `hs project upload` after you created your app object component and the associated configuration files.

In your `my-app-card-hsmeta.json` file, add your app object UID to the `objectTypes` array (e.g., `"app_object_uid"` in this example). Each of the available fields in the `.json` file are detailed in the below.

HubSpot Developers

```
    "name": "My app card",
    "description": "An example description of the card, which lives on contact
    "previewImage": {
      "file": "./preview.png",
      "altText": "A short description of the preview image"
    },
    "entrypoint": "/app/cards/MyCard.jsx",
    "location": "crm.record.tab",
    "objectTypes": ["contacts", "app_object_uid"]
  }
}
```

After you've saved the changes to your `example-card-hsmeta.json` file, run `hs project upload`.

Cards are added automatically to the default view for app objects. If the card doesn't automatically show up, learn how to .

App card *-hsmeta.json fields

Fields marked with * are required.

Field	Type	Description
`uid` *	String	The card's unique identifier. This can be any string, but should meaningfully identify the card. HubSpot will identify the card by this ID so that you can change the card's title without removing historical or stateful data, such as the position on the CRM record.
`type`	String	The type of component, which should be `card` in this case.
`config`	Object	An object containing configuration details.
`name` *	String	The card's title, as displayed in HubSpot's UI.
`description`	String	A description of the card.

HubSpot Developers

›

		`altText` field is a short description of the image.
`entrypoint` *	String	The file path of the card's front-end React code.
`location` *	`crm.record.tab` | `crm.record.sidebar` | `helpdesk.sidebar`	Where the card appears in HubSpot's UI. Learn more about **extension location**.
`objectTypes` *	Array	The types of CRM records that the card will appear on.

App object associations

To enable associations between your app object and other CRM objects, include an `app-object-associations` component directory in the project, along with a `*-hsmeta.json` configuration file for each record type you want to define an association for.

The directory structure and example `*-hsmeta.json` file below provide an example of defining an association between an app object with a `uid` of `car-app-object` and contacts.

```
project-folder/
└── src/
    └── app/
        ├── app-hsmeta.json
        └── app-object-associations/
            └── car-to-contact-hsmeta.json
```

Below are the available configuration options for your app object association details in the corresponding `*-hsmeta.json` file.

HubSpot Developers

```
    "firstObjectType": "car-app-object",
    "secondObjectType": "CONTACT",
    "firstToSecondBaseLimit": 5000,
    "secondToFirstBaseLimit": 100,
    "labels": [
      {
        "name": "product_manager",
        "firstObjectTypeLabel": "Managed Widget",
        "secondObjectTypeLabel": "Product Manager",
        "firstToSecondLabelLimit": 1,
        "secondToFirstLabelLimit": 50
      },
      {
        "name": "quality_inspector",
        "firstObjectTypeLabel": "Inspected Widget",
        "secondObjectTypeLabel": "Quality Inspector"
      },
      {
        "name": "technical_support",
        "firstObjectTypeLabel": "Supported Widget",
        "secondObjectTypeLabel": "Technical Support Rep",
        "firstToSecondLabelLimit": 3
      }
    ]
  }
}
```

Association *-hsmeta.json fields

Fields marked with * are required.

Field	Type	Description
`uid` *	String	An internal unique identifier for the association. Must be globally unique within the project.
`type` *	String	The type of component. Must match the name of the parent folder (`app-object-association`).

HubSpot Developers

›

		`secondObjectType` , `firstToSecondBaseLimit` , `secondToFirstBaseLimit` , and `labels` properties, which are defined in detail in the table rows below.
`firstObjectType`	String	The first object type in the association. Can be an app object UID (e.g., `"car-app-object"`) or the `fullyQualifiedName` of a HubSpot-defined object (e.g., `"CONTACT"`).
`secondObjectType`	String	The second object type in the association. Can be an app object UID (e.g., `"car-app-object"`) or the `fullyQualifiedName` of a HubSpot-defined object (e.g., `"CONTACT"`).
`firstToSecondBaseLimit`	Number	The maximum number of `secondObjectType` records that a single `firstObjectType` record can be associated to. The maximum value is 10,000. Note that if this property is not provided, the corresponding association limit will default to ***Many***.
`secondToFirstBaseLimit`	Number	The maximum number of `firstObjectType` records that a single `secondObjectType` record can be associated to. The maximum value is 10,000. Note that if this property is not provided, the corresponding association limit will default to ***Many***.
`labels`	Array	An array of association labels that describe the relationships between associated records. Each label is an object with the following properties available: `name` * : Unique identifier for the first object type in this association. `firstObjectTypeLabel` * : Display label for the first object type in this association. `secondObjectTypeLabel` * : Display label for the second object type in this association. `firstToSecondLabelLimit` : The maximum number of `secondObjectType` records that a single `firstObjectType` record can be associated with using this label. Cannot be greater than the base limit.

HubSpot Developers

›

number of `firstObjectType` records that a single `secondObjectType` record can be associated with using this label. Cannot be greater than the base limit.

Note that if this property is not provided, the corresponding association limit will default to ***Many***.

Last modified on February 23, 2026

Was this page helpful? Yes No

Get started with app objects
‹ Previous

Agents and tools overview (BETA)
Next ›

HubSpot Developers
Copyright © 2026 HubSpot, Inc

Privacy Policy

Security

Website Accessibility

Manage Cookies